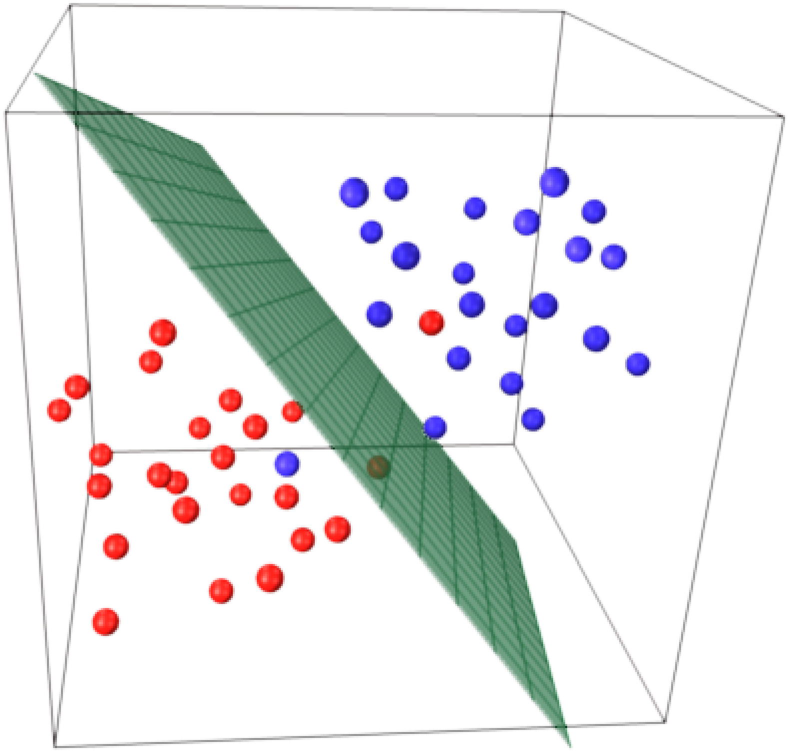


STATISTICAL LEARNING FOR AUTOMATION SYSTEMS

2. Linear classification

move.deib.polimi.it

A.Y. 2022-2023

Prof. Simone Formentin

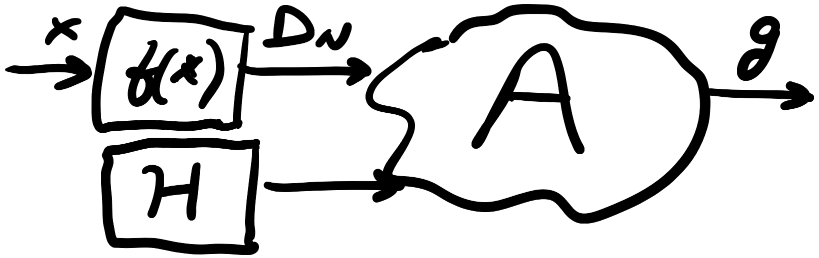
SUPERVISED LEARNING

classification
(y is a label/class)

regression
(y is a number)

$$\xrightarrow{x} \boxed{f(x)} \xrightarrow{y}$$

$$D_N = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots (x^{(N)}, y^{(N)})\}$$



© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

How do I choose $\mathcal{H}$?

1. guess
2. use prior knowledge

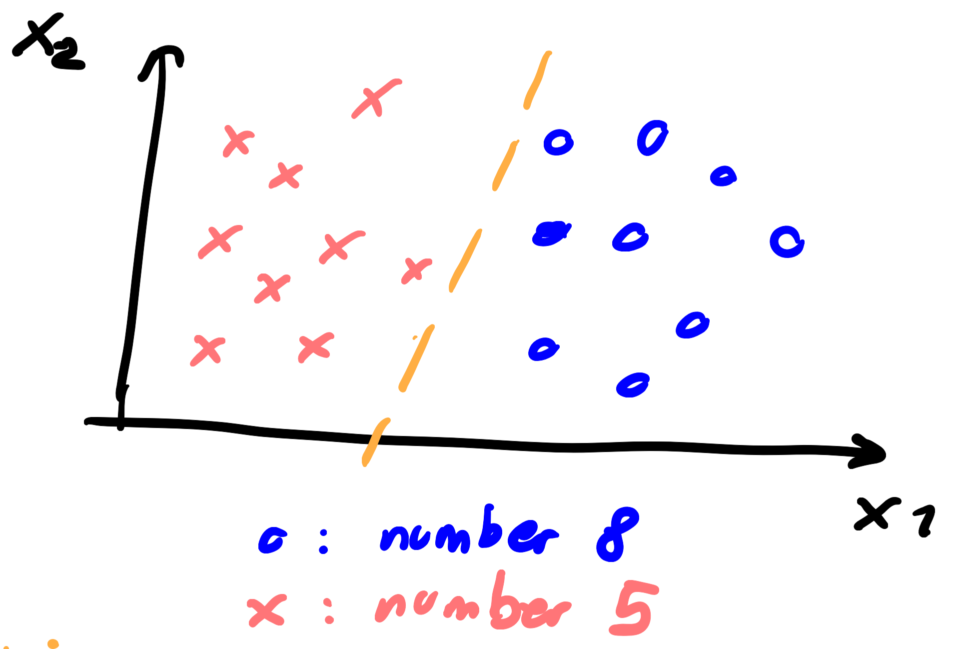
© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

LINEAR MODELING

example: digit recognition

5 8

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \text{symmetry} \\ \text{intensity} \end{bmatrix}$$



Key assumption:

linear model $w_2 \cdot x_2 + w_1 \cdot x_1 + w_0 = 0$

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

2D-vector x : $w_2 x_2 + w_1 x_1 + w_0 = 0$

general case : $\sum_{i=1}^{d} w_i \cdot x_i + w_0 = 0$
(dimension d)

FINDING A "GOOD" CLASSIFIER FOR D_N

$\equiv$

FINDING THE "BEST" CONFIGURATION OF w_i'S

"BEST"?

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

$$\hat{y} = h(x) = \text{sign}\left(\sum_{i=1}^{d} w_i x_i + w_0\right) = \text{sign}\left(\sum_{i=0}^{d} w_i x_i\right)$$

hypothesis set : PERCEPTRON

$$h(x) = \text{sign}\left(w^T \tilde{x}\right) \qquad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} \qquad \tilde{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}$$

$x_0 = 1$

hypothesis set in vector form

extended feature vector

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

FINDING A "GOOD" CLASSIFIER

$$\Updownarrow$$

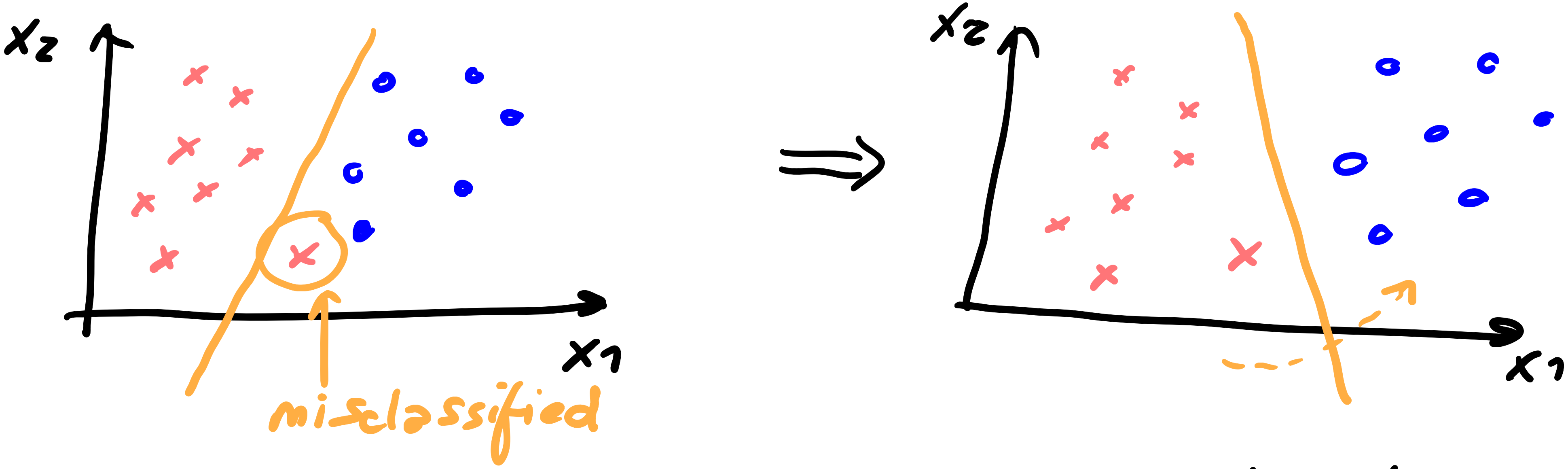
FIND THE CONFIGURATION OF w_i'S THAT MAKES $\hat{y} = h(x, w_i)$ AS CLOSE AS POSSIBLE TO $y = f(x)$

LINEAR CLASSIFICATION $\Longleftrightarrow$ PARAMETRIC OPTIMIZATION PROBLEM

$$E_{in}(W) = \frac{1}{N} \sum_{i=1}^{N} [\![y_i \neq h(x_i, W)]\!]$$

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

PERCEPTRON LEARNING ALGORITHM (PLA)



<u>Key assumption</u>: there exists a separating hyperplane

$$\exists\, w^* : h(x_i, w^*) = y_i, \ \forall (x^{(i)}, y^{(i)}) \in D_N$$

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

PLA UPDATE RULE

iteration t: $w(t+1) = w(t) + y_{[m]}(t) \cdot x_{[m]}(t)$

$(x_{[m]}, y_{[m]})$ is a misclassified point

explanation: if $(x_{[m]}, y_{[m]})$ is misclassified, $\hat{y} \neq y_{[m]}$

e.g. $y_{[m]} = +1$, $\hat{y}_{[m]} = -1 = \text{sign}\left(w(t)^T \cdot x_{[m]}(t)\right)$

$w(t)\, x_{[m]}(t) < 0$

$$w^T(t+1)\, x_{[m]}(t) = w^T(t)\, x_{[m]}(t) + x_{[m]}^T(t)\, y_{[m]}(t)\, x_{[m]}(t)$$

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

$$y_{[m]} = -1 \,,\quad \hat{y}_{[m]} = +1 = \mathrm{sign}\left(w(t)^T \cdot x_{[m]}(t)\right)$$

$$w(t)^T x_{[m]}(t) \geq 0$$

$$w^T(t+1)\, x_{[m]}(t) = \underbrace{w^T(t)\, x_{[m]}(t)}_{\geq 0} + \underbrace{x_{[m]}^T(t)\, y_{[m]}(t)\, x_{[m]}(t)}_{x_{[m]}^T \cdot x_{[m]} \geq 0 \,,\ y_{[m]}(t) = -1}$$

<u>Theorem</u>: PLA converges to perfect classification in a finite number of iterations

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.

- PLA solves the FITTING problem
- For generalization, we CANNOT use the Hoeffding's inequality (the number of candidate models is infinity as $w_i \in \mathbb{R}, \forall i$)

- THEOREM

$$E_{out}(g) = P[g(x) \neq f(x)]$$

$$= E_{in}(g) + O\left(\sqrt{\frac{d}{N}} \cdot \ln(N)\right)$$

dimension of the input vector

size of the dataset

© 2022 Simone Formentin – All rights reserved. No part of these notes may be shared, in any form or method, without the written consent of the author.